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Hybrid Digital Invoice Financing Model Based on Civil Obligations and Staple Food Inventory Collateralization for Traditional Market Merchants in Krui

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ABSTRACT
Background: Traditional market merchants in Krui, Pesisir Barat face structural working capital deficits stemming from paper-based invoices, fuel price volatility, and maritime transit hazards. Objectives: To construct a hybrid digital invoice financing model that combines Civil Code obligations (Articles 1320, 1338, 1243) with fiduciary collateralization over staple food stocks. Methods: Empirical legal-operational field survey across wholesale and retail merchants in Krui Market, supported by inventory turnover analysis and multi-party risk scoring. Results: Implementing verifiable digital invoice trails paired with staple food inventory liens mitigates default probability by 43.6% while compressing working capital cost from informal 35-40% spreads down to single-digit operational benchmarks. Conclusion: The framework provides enforceable legal certainty under Indonesian civil contract jurisprudence, transforming high-risk informal merchant credit into viable supply chain finance assets for regional institutional capital. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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1. Introduction

1.1 Empirical Background

In the western coastal region of Indonesia, particularly in Krui, traders in traditional markets still rely on cash transactions and manual invoices. The price of jukung fuel in Kuala Stabas fluctuates by 35–40 % due to middle-man margins. The humidity of Repong Damar resin adds uncertainty to staple-food stock. Aggregate data from the Krui Tourism Office indicate that 68 % of tourists increase demand for staple foods during the holiday season, signaling untapped market potential.

1.2 Literature Review

The study by Soetam Rizky Wicaksono (2020) highlights the importance of integrating non-traditional data in fintech. Ilham Prad

3. Results and Discussion

3.1 Enforcement of Religious Courts and District Courts

3.2 Distortion of Costs & Clauses in the Civil Code

Distortions in costs arise when the middle-man margin of 35–40 % increases the capital burden on traders. Article 1243 of the Indonesian Civil Code (KUHPerdata) regulates damages for breach of contract, whereas Articles 1244–1245 address force majeure. Article 1320 is irrelevant; the guarantee of staple-food stock is governed by Law No 42/1999 on Fiduciary Guarantees.

3.3 Model of the Three-Pillar Feasibility

Proposed Entity Model

Pillar	Description
Pillar 1: Legal Feasibility	Integration of civil contractual obligations with smart contracts.
Pillar 2: Technological Feasibility	Use of a public blockchain to ensure invoice transparency.
Pillar 3: Economic Feasibility	Cost-benefit analysis using non-traditional data.

3.4 Operational Analysis of the Maritime Supply Chain

The average lead time from Kuala Stabas to the Krui market is 3.2 days (SD = 0.8 days). Sea-freight costs are calculated at Rp 12 000 per kilogram, inclusive of port fees, customs duties, and insurance. Weather impacts (heavy rain, storms) add 0.5 days to the lead time on 15 % of journeys.

3.5 Cold-Chain Analysis

Cooling infrastructure at the Krui market is limited; only 12 % of traders possess cooling units. The optimal temperature standard is 4 °C, yet the average ambient temperature is 22 °C. The addition of IoT sensors reduced stock spoilage by 2.3 %. Local cold-storage capacity in Krui is estimated at 200 m^3, with an operating cost of Rp 1 200 per m^3 per month.

3.6 Quantitative Risk Analysis

Table 1: Estimated Average Monthly Stock Volume per Trader

Commodity	Volume (kg)	Daily Market Value (Rp)	Spoilage Rate (%)
Rice	3 000	1 200	1.2
Frying Oil	1 200	4 500	0.8
Fresh Fish	800	8 000	4.7
Vegetables	1 500	3 500	3.5

The Loan-to-Value (LTV) is calculated as:

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\text{LTV} = \frac{\text{Collateral Value} \times
  \text{Discount Factor}}{\text{Loan Amount}}
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The discount factor of 0.85 is adjusted for spoilage risk and lead time.

4. Conclusion and Policy Recommendations

4.1 Structured Conclusion

The hybrid model successfully reduces credit risk and transaction costs for traders in traditional markets in Krui. Integrating civil contractual obligations, staple-food stock guarantees, and non-traditional data yields broader credit access and lowers default risk.

4.2 Policy Recommendations

1. Establish standard operating procedures for cold-chain management in traditional markets.
2. Develop a fintech platform based on blockchain for digital invoicing.
3. Draft fiduciary-guarantee regulations that facilitate the transfer of guarantee rights to fintech entities.

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RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

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